Ways to Write a 2 by n matrix

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1 Problem 1324

Proposed by Sela Fried, Israel Academic College, Israel. Let n be a positive integer. In how many ways can we write the numbers $1, 2, ..., 2n$ as the elements of a $2 \times n$ matrix such that no two consecutive numbers are in the same column?

2 Solution

There are $n!$ ways to arrange the columns in a $2 \times n$ matrix. Additionally, each column can be flipped independently by swapping its 2 entries, giving 2^n possible combinations of column flips. Therefore, for each set of column pairs, there are $2^n \cdot n!$ possible matrix arrangements. All that remains is to count pairings of integers in columns.

We proceed by the Principle of Inclusion-Exclusion, disregarding order. First, we count all possible pairings of the numbers into columns. There are (2n)! ways to arrange the elements of a 2xn matrix, we proceed to divide by the number of column swaps and flips to give the total number of distinct pairings.

We proceed to subtract cases where at least one pair consists of consecutive integers. However, arrangements containing multiple consecutive pairs are counted repeatedly. To correct this, we add back arrangements containing 2 consecutive pairs, subtract arrangements containing 3 consecutive pairs, add back arrangements containing 4 consecutive pairs, and continue alternating in this manner up to i consecutive pairs.

We determine how many ways there are to choose i disjoint consecutive pairs from the numbers 1,2,..., $2n$. After fixing i consecutive pairs, there are $(2n-2i)$ integers remaining. There are $(2n-2i)!$ permutations of these integers. However, many of these permutations correspond to the same arrangement of columns. For each of the $n-i$ remaining columns , the 2 entries can be swapped, giving 2^{n-i} equivalent permutations. In addition, the $n-i$ pairs can be rearranged among themselves, giving $(n-i)!$ equivalent permutations. Therefore, each unique arrangement of the remaining columns is

$$\frac{(2n-2i)!}{2^{n-i}\cdot(n-i)!}.$$

We have counted the number of ways to arrange the remaining integers after fixing i consecutive pairs. We must now account for the number of ways to choose i disjoint consecutive pairs, denoted by $N(2n,i)$. Combining this with our previous work yields

$$\frac{(2n-2i)!}{2^{n-i}\cdot(n-i)!}\cdot(N(2n,i))$$

matrix arrangements.

Indexing over i pairs yields

$$\sum_{i=0}^{n}(-1)^i\cdot\frac{(2n-2i)!}{2^{n-i}\cdot(n-i)!}\cdot(N(2n,i))$$

We then multiply this summation by the number of matrix arrangements $2^n\cdot n!$

$$2^n\cdot n!\cdot\sum_{i=0}^{n}(-1)^i\cdot\frac{(2n-2i)!}{2^{n-i}\cdot(n-i)!}\cdot(N(2n,i))$$

To construct an explicit formula for $N(2n,i)$, we pick consecutive pairs in a $1\times 2n$ grid of squares with $(2n-2i)$ 1×1 tiles and i 1×2 tiles representing consecutive pairs. There are $2n-2i+i=2n-i$ places each tile can be. We can pick the positions of each tile by choosing the places for the i 1×2 tiles from $2n-i$, and the 1×1 tiles will take the remaining $2n-2i$ spots, yielding:

$$N(2n,i)=\binom{2n-i}{i}$$

This leads to the final count of

$$2^n \cdot n! \cdot \sum_{i=0}^{n} (-1)^i \cdot \frac{(2n-2i)!}{2^{n-i} \cdot (n-i)!} \cdot \binom{2n-i}{i}$$

matrix arrangements.